

Target Detection via Bayesian-Morphological Saliency in High-Resolution SAR Images

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Abstract—The classical target detection methods in synthetic aperture radar (SAR) images are mainly dependent on the intensity differences between the targets and clutter. Although they are effective in the simple scenes with high signal-to-clutter ratio (SCR), they may lose effectiveness in the complex scenes with low SCR. Generally, in high-resolution SAR images, the targets present not only high intensities but also specific size characteristics compared with the clutter. Based on this fact, in this paper, we propose a new target detection method for high-resolution SAR images via Bayesian-morphological saliency, which mainly contains two stages: Bayesian saliency map construction and morphological saliency map construction. The Bayesian saliency map can obtain the complete structures of the bright objects including the targets of interest and some bright clutter, via the superpixel segmentation and Bayesian framework. Furthermore, the morphological saliency map can highlight the targets of interest while suppressing both the natural and man-made clutter via the size prior information of the targets. The experimental results on the miniSAR real data set show that the proposed target detection method is effective.

Index Terms—Bayesian saliency, morphological saliency, saliency detection, superpixel, synthetic aperture radar (SAR), target detection.

I. INTRODUCTION

WITH the quick development of synthetic aperture radar (SAR) imaging technology, the resolution of SAR images becomes higher and higher. Target detection, the first stage of the SAR automatic target recognition [1]–[4], is widely used in the military and civilian fields. However, target detection in high-resolution SAR images still faces some difficulties, especially in the complex scenes, such as urban areas.

The constant false alarm rate (CFAR) [5]–[8], which resorts to the pixel-level intensity differences between the targets and clutter, is commonly used for target detection in SAR images [9]. A widely used CFAR is the two-parameter CFAR [5], [6], which assumes the background clutter obeying the Gaussian distribution. The performance of the two-parameter CFAR is good in some simple scenes with high

signal-to-clutter ratio (SCR), e.g., in situations where a single target is located in the homogeneous background, however, it may suffer from the performance degradation in some complex scenes with low SCR, e.g., in multitarget situations and urban areas. Moreover, the detection performance of CFAR is easily affected by the speckle noise, since CFAR is a pixel-based detection method.

Some studies have been devoted to reduce the speckle noise of SAR images [28], [29]. Yue and Jiang [28] despeckled the SAR images in the nonsubsampling contourlet transform domain based on maximum *a posteriori* and nonlocal constraint. Xu *et al.* [29] filtered the SAR images in the nonlocal principal component analysis domain. Both Yue and Xu denoised the SAR images in a nonlocal manner, which can improve the speckle filtering performance. The nonlocal filters are different from the classical local filters, such as Gaussian filters, in two aspects. First, the nonlocal filters use the patch centered at the pixel rather than the single pixel for filtering, which can include some regional information of the pixel. Second, the nonlocal filters select the homogeneous pixels, i.e., the pixels which are similar to the current pixel to be filtered, for filtering from a nonlocal area, rather than from a local area surrounding the current pixel. Recently, Yu *et al.* [10] proposed a superpixel-based CFAR target detection method for SAR images. Yu *et al.* [10] did not use any speckle noise reduction methods to preprocess the SAR images, instead, they introduced the superpixels into their target detection method, which is robust to the speckle noise.

However, both the conventional CFAR and superpixel-based CFAR have the following disadvantages. First, it is difficult to choose an appropriate clutter statistical distribution for target detection, especially in the complex scenes, and an inappropriate model will lead to a poor detection performance. Second, it is difficult to eliminate the numerous man-made clutter false alarms, such as buildings, with only considering the intensity differences between the targets and the clutter. Third, it is difficult to detect the targets which have low local contrasts but high global contrasts by just using the background clutter surrounding the targets.

All the above-mentioned issues inspire us to develop an effective target detection method for high-resolution SAR images having the following properties.

- 1) *Robust to the Speckle Noise*: The speckle noise makes the SAR image fluctuate and heavily affects the detection performance of the classical target detection methods, such as the pixel-based two-parameter CFAR. An excellent detection algorithm should be robust to the speckle noise.

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- 2) *Free of Clutter Statistical Distribution Hypothesis*: Since the statistical property of the clutter varies with the scenes, it is difficult to determine a fixed clutter statistical distribution model for the target detection method. An excellent detection algorithm should be free of the clutter statistical model.
- 3) *Reducing the Man-Made Clutter False Alarms*: Due to the radar echo intensities of the man-made clutter are similar to or even higher than those of the targets in SAR images, numerous man-made clutter false alarms will be generated if the target detection method only considers the intensity information. Therefore, not only the intensity information but also the other properties of the targets, such as the size information should be used in the new detection method.
- 4) *High Detection Rate of the Targets Which Have Low Local Contrasts But High Global Contrasts*: There usually exists some targets which have low contrasts to their local surroundings but have high global contrasts to its whole backgrounds in SAR images, and these targets are easily to be missed by some local-based methods, such as the two-parameter CFAR. Therefore, the new algorithm should highlight these targets, such as introducing the global contrast.

Saliency detection, as an effective way to imitate the ability of human visual attention system, is widely used for detecting targets in optical images. A large number of saliency detection models for optical images have been proposed [11], which can be divided into the local contrast models, e.g., center-surround model [12] and saliency detection by self-resemblance model [13], and the global contrast models, e.g., spectral residual model [14] and superpixel-level global contrast model [15]. These traditional saliency detectors mainly consider the intensity, color, or orientation difference between the salient object and its surrounding background [11]–[15]. They are usually effective for homogeneous background; however, they are not robust in dealing with heterogeneous background. To solve this problem, some studies introduced the center prior [31], [32], and boundary prior [33], [34], into the saliency detectors. The center prior hypothesizes that image center is likely to contain salient objects, and the boundary prior assumes that regions near image boundaries are likely to be background. In [31]–[34], the saliency detectors with the certain priors were applied to optical images with heterogeneous backgrounds.

Although the traditional saliency detectors are effective for optical images, they cannot be directly used in SAR images, due to the following characteristics of SAR images. First, the SAR image usually has heterogeneous background, such as man-made clutter. Second, the targets of interest may appear anywhere in the SAR image, thus the center prior and boundary prior cannot be used in SAR images. Based on the characteristics of SAR images, some saliency detection methods for SAR images have been proposed. These methods can be divided into the data-driven methods [16], [17], [30], and the task-driven methods [18], [22]. The data-driven methods mainly consider the intensity information of the targets, e.g., pattern recurrence (PR) [30] used patch-level instead of pixel-level intensity contrast to highlight the targets while

overcoming the speckle noise. The task-driven methods consider not only the intensity information but also some prior knowledge of the targets to be detected, e.g., the ship targets are usually in water regions rather than land [18]. In general, due to the utilization of the prior information of the targets to be detected, task-driven methods yield better performance than the data-driven methods in heterogeneous background. However, which prior information should be introduced and how to introduce the prior information need to be further researched. In this paper, we devote to develop a new saliency detector which is robust for SAR images with heterogeneous background.

Recently, Tong *et al.* [19] proposed the Bayesian saliency detection model. Within the Bayesian framework, this model combines the prior map obtained by the superpixel-level local contrast method with the two likelihood maps obtained by the convex hull, which can uniformly highlight the salient regions and suppress the nonsalient regions. However, this method cannot directly be applied to the SAR images. First, since the pixels of SAR images are very discrete and the scenes are usually very large, it is difficult to obtain the convex hull which includes the foreground like optical images do. Second, due to the lack utilization of the prior information of the targets, it is inevitable to highlight all the bright objects including the targets of interest and man-made clutter.

In the field of optical remote sensing, Huang and Zhang [20], [21], recently proposed a new building extraction method based on differential morphological profiles (DMPs), which can be seen as a saliency detection model, and in this paper we refer to it as morphological saliency. This method can highlight the buildings while suppressing the backgrounds by means of multidirectional and multiscale morphological filters in the original image. The buildings are smooth and connected enough in optical remote sensing images, thus the morphological filters can be effective in extracting the structural information of the buildings. However, due to the speckle noise, the intensities of the objects in SAR images are fluctuant, making it difficult to extract the desired targets via the morphological saliency from the original SAR images.

According to the above analysis, in this paper, we propose a novel target detection method for high-resolution SAR images via Bayesian-morphological saliency (BMS), which mainly contains two stages: Bayesian saliency map construction and morphological saliency map construction. The first stage is used to obtain the smooth and complete structures of the bright objects, including the targets of interest and some bright clutter. If there are not many bright clutter in the SAR image, the first stage usually can obtain good detection performance; otherwise, the second stage is employed to further exclude the bright clutter. The proposed BMS method is specially designed to deal with the target detection problems in SAR images with heterogeneous background and it has the following properties.

First, we use the superpixel instead of the pixel as the basic operation unit to construct the prior map for Bayesian saliency map, which is more robust to the speckle noise. Second, we do not assume any statistical distribution models for the prior map and two likelihood maps, although the Bayesian framework is employed to calculate the Bayesian saliency. Here, the prior

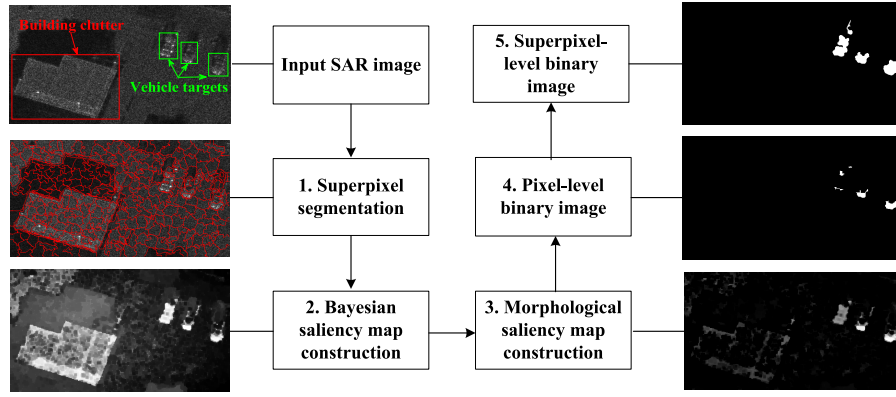


Fig. 1. Flowchart of the proposed target detection method for high-resolution SAR images.

map is constructed based on global contrast in the SAR image, and the two likelihood maps are generated based on the histograms of the preliminary foreground and background of the SAR image. Third, we construct the prior map for the Bayesian saliency in a global contrast manner rather than a local contrast manner. Compared with some local contrast-based methods, e.g., the original Bayesian saliency and two-parameter CFAR, the proposed BMS method can improve the detection rate of the targets, which have low local contrasts but high global contrasts. Finally, we introduce the morphological saliency into our target detection framework for SAR images to exclude the man-made clutter, since the Bayesian saliency map can uniformly highlight not only the targets but also the man-made clutter. Although the targets of interest, e.g., vehicle targets, and man-made clutter (heterogeneous background), e.g., buildings, usually have similar intensities in the Bayesian saliency map or original SAR images, they are usually much different in size. In our proposed BMS method, we employ the size prior information of the targets of interest via the morphological saliency. The morphological saliency map for SAR image is constructed from the Bayesian saliency map rather than from the original SAR image. Compared with the original SAR image which is fluctuate, the bright objects are smoother and their structures are more complete in the Bayesian saliency map. Then the size prior information can be employed via the morphological saliency based on the Bayesian saliency map to extract the targets of interest and reduce the man-made clutter.

The remainder of this paper is organized as follows. Section II introduces the proposed target detection method. Section III gives the experimental results and analysis based on the miniSAR real data set. Finally, Section IV concludes this paper.

II. PROPOSED TARGET DETECTION METHOD

A. Framework of the Proposed Method

The flowchart of the proposed target detection method for high-resolution SAR images is shown in Fig. 1. A subimage example cut from a miniSAR image is shown along each step for the better illustration purpose. In the first image, the three vehicle targets of interest in this example are marked by the green boxes and the building clutter is marked by the red

box. Here, both the vehicle targets and the building clutter are with high intensities; therefore it is difficult to detect the vehicle targets from this subimage by only using intensity information. Fig. 1 qualitatively shows that the proposed target detection method, which considers both the intensity and size information of the targets of interest, can not only detect the vehicle targets but also exclude the building clutter. A brief explanation for each step is given in the following.

Step 1: The input SAR image is segmented into the superpixels, which are small regions obtained by over segmentation. There are two advantages of introducing the superpixels into our detection method. First, we use each superpixel instead of each pixel as the elementary unit to construct the prior map for the Bayesian saliency map, which is more robust to the speckle noise. Second, the clustering procedure followed by the target detection procedure can utilize the segmented superpixels instead of the original pixels as the elementary units, and the clustering result based on the superpixels is more accurate than that based on the pixels [10], since the superpixels can partially preserve the shapes of the targets. Here the superpixel segmentation algorithm proposed by Yu *et al.* [10] is adopted in this paper, where they introduced the generalized likelihood ratio dissimilarity measurement into simple linear iterative clustering [23] to make the algorithm more suitable for SAR images. In this example, the generated superpixels are marked by the red curves.

Step 2: The purpose of the Bayesian saliency map is to highlight all the bright objects and obtain their complete structures via the superpixel segmentation and Bayesian framework. From this example, we can see that the Bayesian saliency map can uniformly highlight the vehicle targets and building clutter. The details about construction of the Bayesian saliency map will be further introduced in Section II-B.

Step 3: The purpose of the morphological saliency map is to retain the targets while suppressing both the natural and man-made clutter in the Bayesian saliency map. In this example, the vehicle targets are prominent and the building and the other clutter are rather weak. The details about construction of the morphological saliency map will be further introduced in Section II-C.

Step 4: Since the intensities of the targets are much higher than those of the clutter in the morphological saliency map,

we can use a global threshold to extract the targets from this map. In this example, the vehicle targets are the white pixels in the segmented binary map. The selection of the global threshold will be discussed in Section II-D.

Step 5: In this step, we get the superpixel-level detection result based on the pixel-level binary image obtained in Step 4, since the superpixels can partially preserve the shapes of the targets. For each superpixel, if the ratio of the number of the detected pixels inside the superpixel to the total number of pixels in the superpixel is greater than a threshold T_R , it is regarded as a detected superpixel [10]. $T_R = 3\%$ in Fig. 1 and following experiments.

Among the above steps, Steps 2–4 are crucial for the proposed target detection method, and we will explain their details in the following.

B. Bayesian Saliency for SAR Image

1) Prior Map: Assume that an input SAR image is segmented into superpixels $\{s_i\}_{i=1}^N$, where N is the number of superpixels. For any superpixel s_i , the prior saliency is defined as

$$P(s_i) = \sum_{s_k \neq s_i} w_{\text{size}}(s_k) w_{\text{spatial}}(s_k, s_i) D_{\text{intensity}}(s_k, s_i) \quad (1)$$

$$w_{\text{size}}(s_k) = A(s_k) \quad (2)$$

$$w_{\text{spatial}}(s_k, s_i) = \exp\left(-\frac{D_{\text{spatial}}(s_k, s_i)}{\sigma_{\text{spatial}}^2}\right) \quad (3)$$

where $w_{\text{size}}(s_k)$ is the size weight, $w_{\text{spatial}}(s_k, s_i)$ is the spatial distance weight, $D_{\text{intensity}}(s_k, s_i)$ represents the Euclidean distance between the intensity mean values of superpixels s_k and s_i , $A(s_k)$ is the number of pixels in superpixel s_k , $D_{\text{spatial}}(s_k, s_i)$ indicates the spatial Euclidean distance between the centers of superpixels s_k and s_i , and σ_{spatial} controls the strength of spatial distance weighting. In fact, $D_{\text{intensity}}(s_k, s_i)$ is the contrast between the superpixel s_k and the superpixel s_i .

Compared with using the weighted sum of the current superpixel's contrasts only to its neighborhood superpixels to define the prior saliency for each superpixel in [19] (local contrast), (1) essentially defines the prior saliency for each superpixel as the weighted sum of the current superpixel's contrasts to all other superpixels in the whole SAR image (global contrast). For a smaller σ_{spatial} , the prior saliency of the current superpixel is mainly dependent on the superpixels close to the current superpixels; while for a larger σ_{spatial} , the superpixels far from the current superpixel also bring much contribution to the prior saliency of the current superpixel. In Fig. 1 and following experiments, we use $\sigma_{\text{spatial}}^2 = 0.4$ with pixel coordinates normalized to $[0, 1]$, same as the corresponding setting used in [15].

For the saliency detection method, a region with a larger saliency value means higher probability to be a potential target region (foreground). Therefore, we normalize (1) as

$$\bar{P}(s_i) = \frac{P(s_i)}{\sum_{i=1}^N P(s_i)}. \quad (4)$$

Here the normalized prior saliency $\bar{P}(s_i)$ denotes the prior probability of the superpixel s_i belonging to the foreground, while $1 - \bar{P}(s_i)$ describes the prior probability of the superpixel s_i belonging to the background (objects not of interest, e.g., the natural and man-made clutter).

2) Likelihood Maps: Based on the preliminary foreground and background obtained by the multiscale target detection method in [22], we can obtain the intensity histograms of the foreground region and background region, H^1 and H^0 , respectively. Then the observation likelihood of the pixel z in the SAR image is defined as

$$L(z|S^1) = \frac{N^1(z)}{N_{S^1}} \quad (5)$$

$$L(z|S^0) = \frac{N^0(z)}{N_{S^0}} \quad (6)$$

where S^1 and S^0 are used to denote the foreground and background, respectively, $N^1(z)$ and $N^0(z)$ represent the numbers of pixels with the intensity value the same as z in H^1 and H^0 , respectively, and N_{S^1} and N_{S^0} denote the total pixel numbers of the foreground and background, respectively. $L(z|S^1)$ indicates the likelihood of the pixel z belonging to the foreground; while $L(z|S^0)$ indicates the likelihood of the pixel z belonging to the background.

3) Bayesian Saliency Map: We combine the prior map with the likelihood maps within the Bayesian framework to obtain the Bayesian saliency map. Note that the prior map in (1) is defined at the superpixel-level, whereas the likelihood maps in (5) and (6) are defined at the pixel-level. In order to integrate the prior map and the likelihood maps, the prior saliency of a pixel is set to be equal to that of the superpixel which contains the pixel. Then the Bayesian saliency of the pixel z is defined as

$$B(z) = \frac{\bar{P}(z)L(z|S^1)}{\bar{P}(z)L(z|S^1) + (1 - \bar{P}(z))L(z|S^0)} \quad (7)$$

where $\bar{P}(z) = \bar{P}(s_i)$ with the pixel z belonging to the superpixel s_i . Here $\bar{P}(s_i)$, $L(z|S^1)$, and $L(z|S^0)$ are calculated with (4)–(6), respectively.

C. Morphological Saliency for SAR Image

The morphological saliency map for an SAR image is constructed based on the DMPs from its Bayesian saliency map, and each DMP is the difference between two morphological profiles (MPs) which are generated by using the opening-by-reconstruction morphological operation on the Bayesian saliency map with a series of linear (line) structural elements (SEs). The details about the construction of the morphological saliency map are introduced below.

1) MP: The MP is defined as

$$\text{MP}(\text{dir}, L) = \text{open}^{\text{re}}(\text{dir}, L, B) \quad (8)$$

where $\text{open}^{\text{re}}(\cdot, \cdot, \cdot)$ is the opening-by-reconstruction [24] morphological filter, dir and L denote the direction and the length of a linear SE, respectively, B is the Bayesian saliency map of an SAR image, and $\text{MP}(\text{dir}, L)$ is the MP of the opening-by-reconstruction with a linear SE along the direction dir and of the length L .

Mathematical morphology is an effective tool for extracting image components that are useful in the description of region size and shape [24]. For morphological image processing, the small sets, which are also called SEs, are used to probe an image for properties of objects of interest. The SEs should be selected according to the prior information (size or shape) of the targets to be detected. In SAR images, the targets of interest, such as vehicles, usually have a limited size range and may locate in an image along any directions. In this paper, we adopt the linear SEs for detecting the targets in SAR images. The linear SE considers both the size information and directional information, and therefore the linear SE is appropriate for detecting the targets with a limited size range and along different directions from the cluttered SAR images.

In (8), the opening-by-reconstruction morphological filter is used to preserve the bright objects with the lengths equal to or larger than L and remove the bright objects with the lengths smaller than L along the direction dir . Note that here the reconstruction filter is used since it has been proven to have a better shape preservation than classical morphological filters [24].

2) *DMP*: The DMP along direction dir is calculated as

$$DMP(dir) = |MP(dir, L_{max}) - MP(dir, L_{min})| \quad (9)$$

where L_{max} is the maximum length of the targets to be detected, L_{min} is the minimum length of the targets to be detected, $MP(dir, L_{max})$ is the MP with a direction of dir and a scale of L_{max} , and $MP(dir, L_{min})$ is the MP with a direction of dir and a scale of L_{min} . Note that $MP(dir, L_{max})$ can preserve the bright objects with a length equal to or larger than L_{max} along the direction dir , and $MP(dir, L_{min})$ can preserve the bright objects with a length equal to or larger than L_{min} along the direction dir , therefore the difference between $MP(dir, L_{max})$ and $MP(dir, L_{min})$ can preserve the bright objects whose lengths are between L_{min} and L_{max} while removing the other bright objects along the direction dir .

3) *Morphological Saliency Map*: The morphological saliency map is calculated as

$$M_s = \frac{\sum_{dir} DMP(dir)}{D} \quad (10)$$

where D denotes the number of the DMPs along different directions. Since the targets may locate along any directions in an SAR image, the combination of the DMPs with different directions $\{DMP(dir)\}$ makes the morphological saliency map M_s more robust.

4) *Simulated Example*: As mentioned above, the morphological saliency map obtained by the combination of the DMPs with different directions can highlight the targets with the sizes between L_{min} and L_{max} while removing the other bright objects whose sizes are not in this range. In the following, we use a simulated example shown in Fig. 2 to further illustrate the effect of the morphological saliency map. The size of the original simulated image is 200×200 pixels. There are three objects in the original image, the rectangular object A with 12×12 pixels, the rectangular object B with 30×20 pixels and the semicircular object C with its radius of 30 pixels. Note that the objects A and B have the same intensity, and

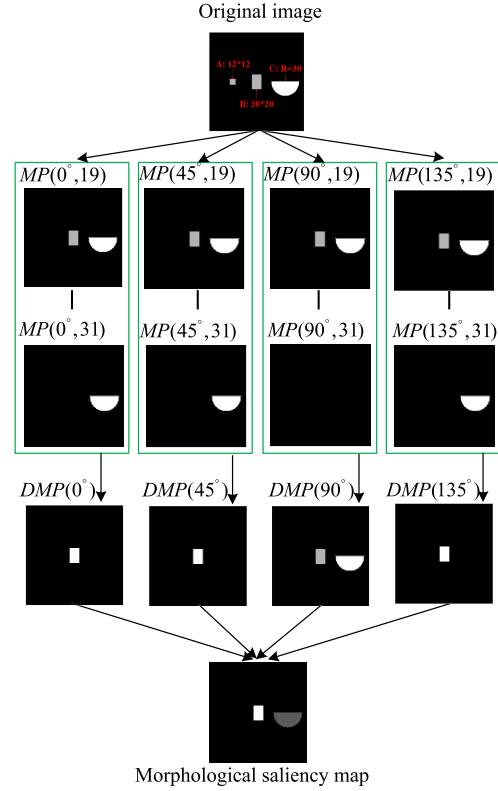


Fig. 2. Example of morphological saliency map construction on a simulated image.

both of them have lower intensity than the object C in the original image. Here, we assume B is the target of interest, A and C are the objects not of interest. In this example, $dir \in \{0^\circ, 45^\circ, 90^\circ, 145^\circ\}$, L_{min} is set to be 19 and L_{max} is set to be 31.

As shown in Fig. 2, B is highlighted, A is entirely removed, and C is highly suppressed in the morphological saliency map. Since all the lengths of A along the four directions are not in the range of between L_{min} and L_{max} , A is removed in the morphological saliency map; the length of C along 90° is within this range, and the lengths of C along the other directions are not in this range, therefore C is highly suppressed rather than entirely removed; the length of B is within the above range along all directions, and B is much brighter than C in the morphological saliency map. Then, based on the morphological saliency map, we can extract B easily using a global threshold.

In this paper, we refer to the morphological saliency map obtained from the Bayesian saliency map as BMS map. In Fig. 1 and following experiments, four directions are considered, i.e., $dir \in \{0^\circ, 45^\circ, 90^\circ, 145^\circ\}$. In our proposed BMS method, L_{max} is used to eliminate the clutter larger than the targets to be detected and L_{min} is used to eliminate the clutter smaller than the targets to be detected. Thus L_{max} should be set according to the maximum length of the target to be detected, since the length of target in the SAR image is usually smaller than that of the real target of interest; L_{min} should be smaller than L_{max} and larger than the possible small clutter.

TABLE I
COMPUTATIONAL COMPLEXITY OF THE BMS

BMS algorithm	Supapixel segmentation	Bayesian saliency map construction	Morphological saliency map construction
Computational complexity	$o(M \times Q \times I)$	$o(N^2)$	$o(M \times (L_{\min} + L_{\max}))$

In Fig. 1 and following experiments, the employed miniSAR images have resolutions of $0.1 \text{ m} \times 0.1 \text{ m}$, where the vehicles are the targets of interest. Here the maximum length of the vehicle targets is about 4 m. According to the resolution of the SAR image, a vehicle target to be detected has a maximum length of about 40 pixels, and thus $L_{\max}$ is set to 40. We set $L_{\min}$ to be 4 in Fig. 1 and following experiments. In our experiments, we find that the proposed BMS method can obtain satisfactory and robust detection results when $L_{\min}$ is set to be 3–7 for detecting vehicle targets in the miniSAR images.

D. Generation of the Binary Image

The intensities of the targets are usually higher than those of the clutter in the BMS map, thus we can use a global threshold to generate the binary image for target detection

$$\text{Bin}(i, j) = \begin{cases} 1, & \text{if } I_{\text{BMS}}(i, j) \geq T \\ 0, & \text{otherwise.} \end{cases} \quad (11)$$

where $I_{\text{BMS}}(i, j)$ is the intensity value of the BMS map at point (i, j) , T is the global threshold, $\text{Bin}(i, j) = 1$ indicates the point is a target pixel, and $\text{Bin}(i, j) = 0$ represents the point is a nontarget pixel.

Let F be the cumulative distribution function obtained from the intensity histogram of the BMS map. Then T can be obtained from [7]

$$1 - F(T) = \alpha \quad (12)$$

where $\alpha \in [0, 1]$ is an empirical value that indicates the confidence of a pixel being a target pixel, since the targets of interest are highlighted in the BMS map. Moreover, due to the targets of interest, such as the vehicles in Fig. 1, only contain a small proportion of pixels in the whole SAR image, α should be small. Here $\alpha = 0.5\%$ in Fig. 1.

E. Computational Complexity Analysis

The proposed BMS algorithm mainly consists of three steps, i.e., superpixel segmentation, Bayesian saliency map construction, and morphological saliency map construction. Table I gives the computational complexities of the three steps in the proposed BMS algorithm. As shown in Table I, the computational complexity of superpixel segmentation is $o(M \times Q \times I)$, where M is the number of pixels in the SAR image, Q is the number of pixels of the local patch centered at the pixel, and I is the number of iterations; the computational complexity of the Bayesian saliency map construction is mainly from the global contrast in the prior map,

and it is $o(N^2)$ where N is the number of superpixels in the SAR image; the computational complexity of morphological saliency map construction is $o(M \times (L_{\min} + L_{\max}))$, where $L_{\max}$ is the maximum length of the targets to be detected and $L_{\min}$ is the minimum length of the targets to be detected.

In our experiments, Q is set to be 5×5 and I is set to be 5, the same as the corresponding setting used in [12], which usually can obtain satisfactory superpixel segmentation results. The number of superpixels N is usually much smaller than that of the pixels M in the SAR image. In our experiments, N is 9621 with an SAR image having a size of $M = 1638 \times 2510$ pixels, i.e., $N \approx M/400$.

III. EXPERIMENTAL RESULTS AND ANALYSIS

The miniSAR real data set is acquired by Sandia National Laboratories of America.¹ The SAR images in the miniSAR real data set are with complex scenes, where there are vehicles, buildings, roads, grasslands, trees, and so on. We select two typical SAR images with complex scenes from the miniSAR real data set for our experiments, and we refer to them as Image I and Image II, respectively. Their resolutions are $0.1 \text{ m} \times 0.1 \text{ m}$ and their sizes are 1638×2510 pixels. Fig. 3 shows the two original SAR images and their superpixel-level ground truths, where the vehicles are the targets of interest and the others are regarded as the clutter. As we can see from Fig. 3(a) and (b), the building man-made clutter which are marked by red boxes have similar to or even higher intensities than the vehicle targets. Therefore, it is difficult to extract the vehicle targets without generating numerous clutter false alarms, by only considering the intensity information of the targets like the classical target detection methods do. Fig. 4 shows the segmented superpixels in SAR images, and here in order to show clearly we only give the segmentation results for a part of the original images.

The experiments are conducted using a personal computer with Intel Core i5-4590 CPU of 3.3 GHz and memory of 8 GB. The codes are written in MATLAB R2012a.

A. Saliency Map Results

Fig. 5 shows the Bayesian saliency maps of Image I and Image II. As shown in Fig. 5, all the bright objects, including the vehicle targets, the buildings, and the trees, are smoother and their structures are more complete in the Bayesian saliency maps than those in Fig. 3(a) and (b) which are the original SAR images.

Fig. 6 shows the BMS maps of Image I and Image II, from which we can see that the vehicle targets are much brighter than most clutter, e.g., trees, buildings, and therefore we can extract the vehicle targets easily from the BMS map without generating a large number of false alarms. In the following, we will quantitatively compare the proposed target detection method with the traditional target detection methods at the superpixel-level.

¹<http://www.sandia.gov/radar/imagery>

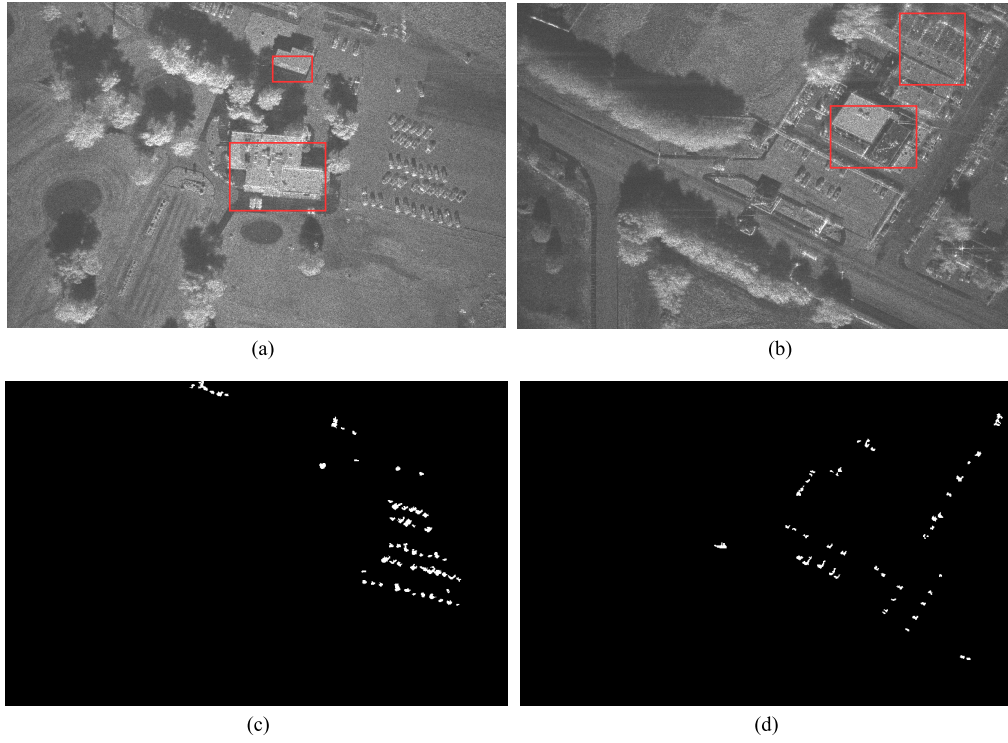


Fig. 3. Original SAR images and their ground truths. (a) and (b) Image I and Image II, respectively, where the vehicles are the targets of interest, and the building clutter marked by red boxes are difficult to eliminate by the classical target detection methods. (c) and (d) Superpixel-level ground truths of the vehicle targets in (a) and (b), marked by the white regions, manually.

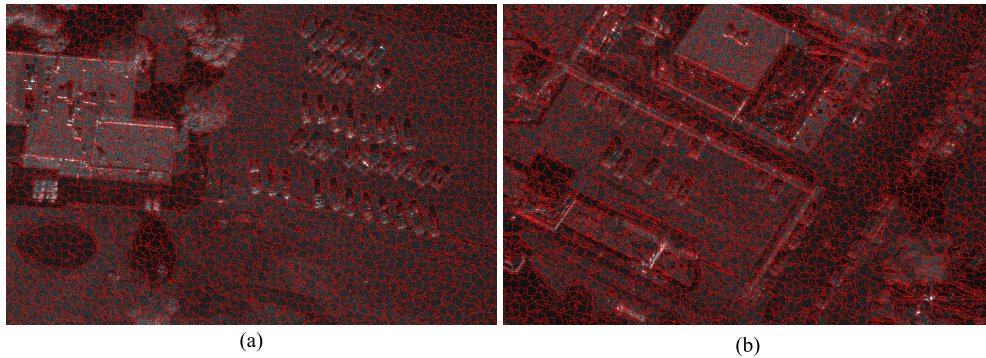


Fig. 4. Segmented superpixels in SAR images. (a) and (b) Segmented superpixels of Image I and Image II, respectively, and here in order to show clearly we only give the segmentation results for a part of the original images.

B. Detection Results

In the following, we compare the proposed BMS method with Local-CFAR [5], [6], Global-CFAR [25], Superpixel-CFAR [10], PR [30], Bayesian saliency for SAR images (BS), and morphological saliency based on the original SAR image (MS). Here Local-CFAR, Global-CFAR, and Superpixel-CFAR are CFAR-based detectors, PR, BS, MS, and BMS are saliency-based detectors. The receiver operating characteristic (ROC) curve [26] and the area under the ROC curve (AUC) [27] are employed for evaluation.

Assume that the ground truth of the target superpixels are denoted as $\{S_T(i)\}_{i=1}^{N_{TS}}$ with N_{TS} being the total number of the target superpixels, and the ground truth of the clutter superpixels are represented as $\{S_C(i)\}_{i=1}^{N_{CS}}$ with N_{CS} being the

total number of the clutter superpixels. Then the probability of detection (PD) and probability of false alarm (PFA) are defined as

$$PD = \frac{N_{DT}}{N_{TS}} \quad (13)$$

$$PFA = \frac{N_{DC}}{N_{CS}} \quad (14)$$

where N_{DT} is the number of the detected target superpixels and N_{DC} is the number of the detected clutter superpixels.

Fig. 7 shows the ROC curves of different target detection methods at the superpixel-level, and Table II gives their corresponding AUCs. The closer to the upper left the ROC curve is, the better its performance is. As shown in Fig. 7, the ROC curves obtained by the proposed BMS method

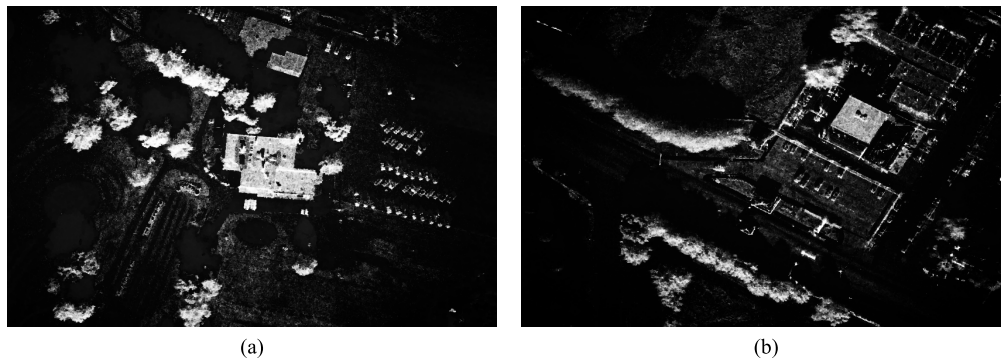


Fig. 5. Bayesian saliency maps of SAR images. (a) Image I. (b) Image II.

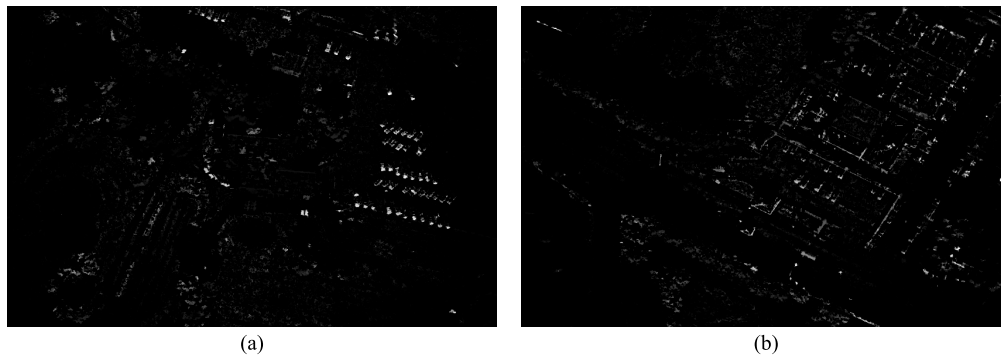


Fig. 6. BMS maps of SAR images. (a) Image I. (b) Image II.

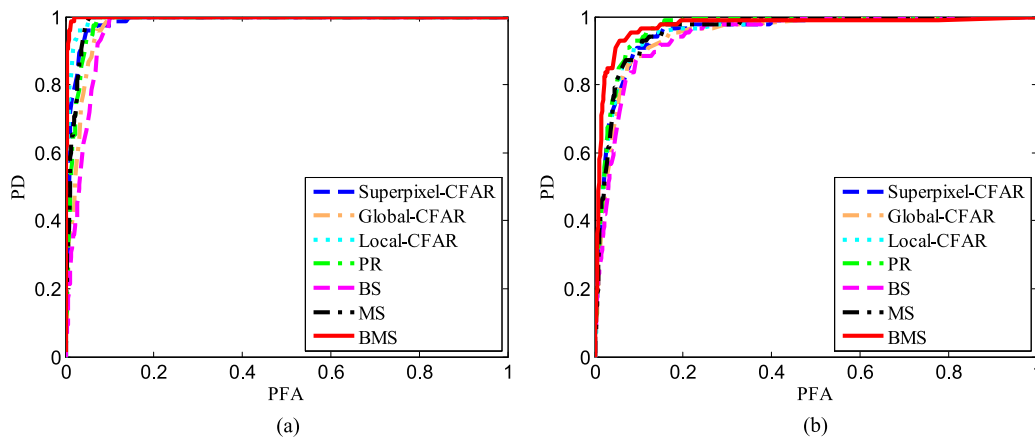


Fig. 7. ROC curves of different target detection methods at the superpixel-level. (a) Image I. (b) Image II. Here Superpixel-CFAR is the superpixel-based CFAR [10], Global-CFAR is the global two-parameter CFAR [25], Local-CFAR is the local two-parameter CFAR [6], PR is the pattern recurrence saliency detector [30], BS is the detection method only using the Bayesian saliency, MS is the detection method only using the morphological saliency based on the original SAR image, and BMS is the proposed detection method in this paper, which combines the Bayesian saliency and the morphological saliency.

TABLE II
AUCs OF DIFFERENT TARGET DETECTION METHODS AT THE SUPERPIXEL-LEVEL

Image	Superpixel-CFAR	Global-CFAR	Local-CFAR	PR	BS	MS	BMS
Image I	0.9850	0.9744	0.9927	0.9823	0.9645	0.9855	0.9975
Image II	0.9572	0.9504	0.9580	0.9684	0.9460	0.9610	0.9710

are better than those of the other target detection methods. The larger the AUC is, the better its performance is. From Table II, we can observe that the detection performance of the

proposed BMS method is better than those of the traditional target detection methods, i.e., Local-CFAR, Global-CFAR, Superpixel-CFAR, PR; meanwhile the proposed BMS method

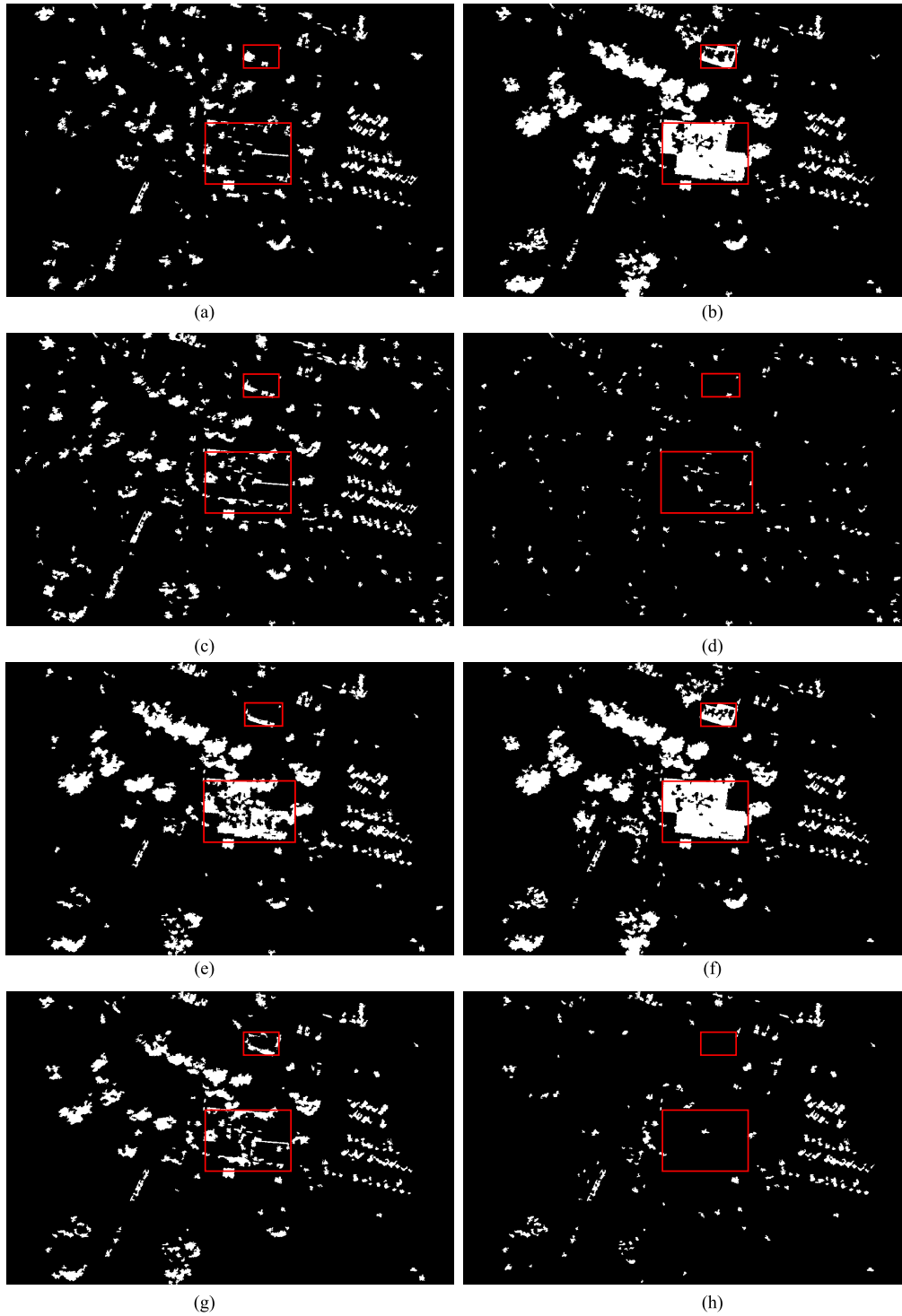


Fig. 8. Detection binary maps of Image I at the superpixel-level. (a) Superpixel-CFAR. (b) Global-CFAR. (c) Local-CFAR. (d) Local-CFAR- Th_{size} . (e) PR. (f) BS. (g) MS. (h) BMS. In these maps, the regions marked by the red boxes indicate the positions of the building clutter in Image I.

which combines the BS and MS can obtain better performance than those using the BS and MS separately.

The optimal operation point is the operation point that makes the following loss function minimum:

$$T_{\text{loss}} = \text{PFA} + \beta(1 - \text{PD}) \quad (15)$$

where β specifies a relative importance of PFA (false alarms loss) versus $(1 - \text{PD})$ (missed targets loss). In this experiment,

$\beta = 1$, which regards PFA and $(1 - \text{PD})$ being of the same importance. Table III gives the optimal operation points of different detection methods. Note that here we also give the optimal operation point of the Local-CFAR detector with size thresholding (Local-CFAR- Th_{size}), i.e., we apply the same size threshold, $L_{\min}$ and $L_{\max}$, as BMS to the detection binary map at the optimal operation points of Local-CFAR. As shown in Table III, the proposed BMS method has lower

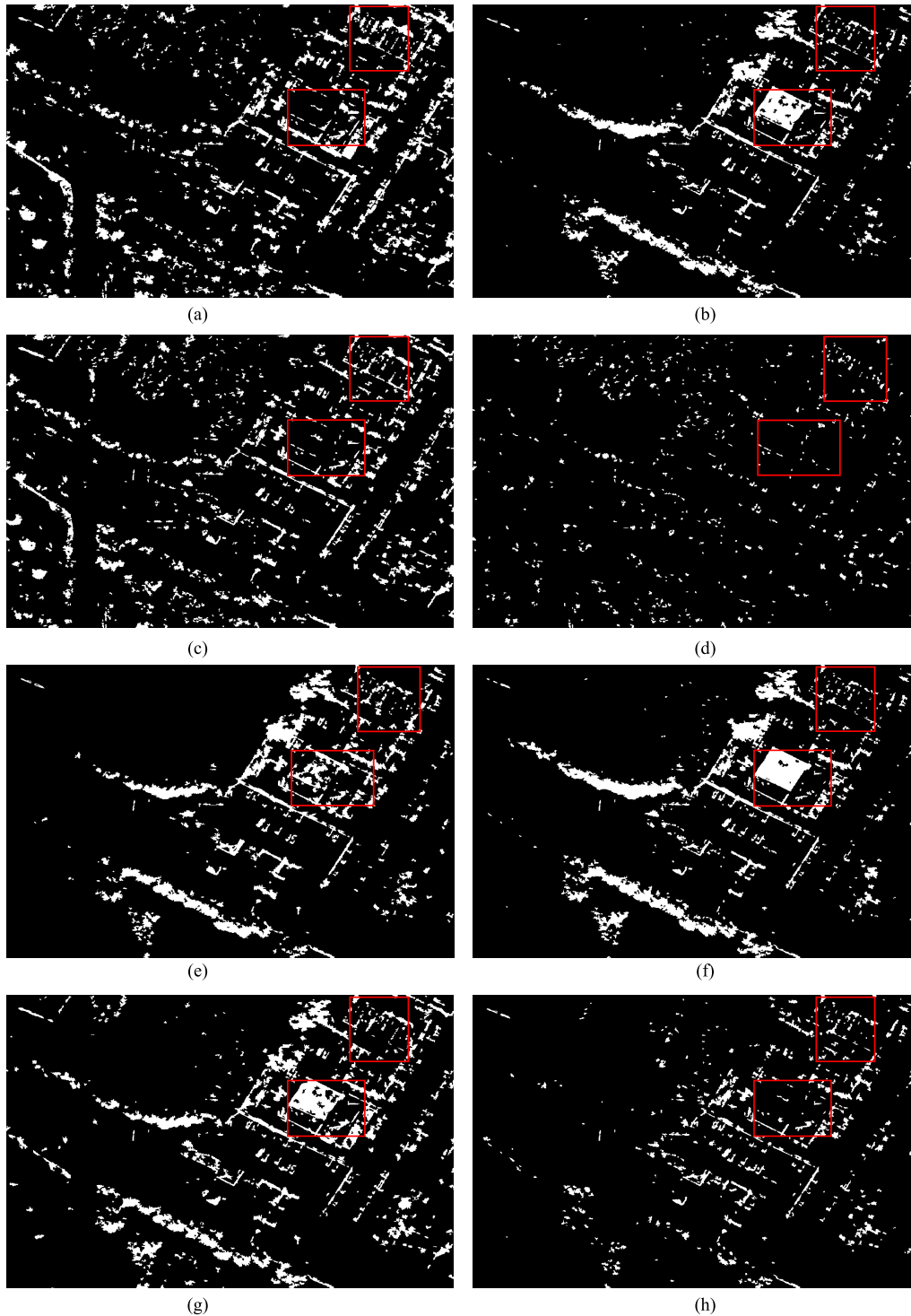


Fig. 9. Detection binary maps of Image II at the superpixel-level. (a) Superpixel-CFAR. (b) Global-CFAR. (c) Local-CFAR. (d) Local-CFAR- Th_{size} . (e) PR. (f) BS. (g) MS. (h) BMS. In these maps, the regions marked by the red boxes indicate the positions of the building clutter in Image II.

PFAs and T_{loss} than the Superpixel-CFAR, Global-CFAR, Local-CFAR, PR, BS and MS, and they have similar PDs. Moreover, the proposed BMS method has much higher PDs and lower T_{loss} than the Local-CFAR- Th_{size} , and they have similar PFAs.

Figs. 8(a)–(c) and 9(a)–(c), and 9(a)–(c) and 9(e)–(h) show the superpixel-level detection binary maps at the optimal operation points of the Superpixel-CFAR, Global-CFAR, Local-CFAR, PR, BS, MS, and the proposed BMS method, where the

regions marked by the red boxes are the regions of the building clutter in Image I and Image II. From Figs. 8(a)–(c) and 9(a)–(c), and 9(e)–(h), we can observe that the proposed BMS target detection method can remove much more building clutter than the other target detection methods, owing to the utilization of the size prior information, i.e., the building clutter have much larger sizes than the vehicle targets.

Figs. 8(d) and 9(d) give the binary maps of Local-CFAR- Th_{size} . Comparing Figs. 8(c) and 9(c) with

TABLE III
OPTIMAL OPERATION POINTS OF DIFFERENT TARGET DETECTION METHODS

Image		Superpixel-CFAR	Global-CFAR	Local-CFAR	Local-CFAR- Th_{size}	PR	BS	MS	BMS
Image I	PD	0.9600	1	1	0.2000	1	1	1	1
	PFA	0.0470	0.1011	0.0614	0.0201	0.0758	0.1011	0.0545	0.0203
	T_{loss}	0.0870	0.1011	0.0614	0.8201	0.0758	0.1011	0.0545	0.0203
Image II	PD	0.9419	0.8837	0.9419	0.3837	0.9302	0.8837	0.9302	0.9302
	PFA	0.1288	0.0897	0.1134	0.0461	0.0864	0.0985	0.1083	0.0569
	T_{loss}	0.1869	0.2060	0.1715	0.6624	0.1562	0.2147	0.1780	0.1267

Figs. 8(d) and 9(d), we can see that numerous false alarms whose lengths are too small or too large are eliminated by Local-CFAR- Th_{size} ; meanwhile, a large number of vehicle targets are also removed together with the false alarms. From Figs. 8(h) and 9(h), we can see that the proposed BMS method can detect most vehicle targets without generating too many false alarms. A main difference between Local-CFAR- Th_{size} and proposed BMS method is the way how to use the size threshold. For Local-CFAR- Th_{size} , the size threshold is applied to the binary map obtained by Local-CFAR. The sizes of the targets in the binary map may be much different from their real sizes, since the completeness and the connectivity of the targets in the binary maps vary with the detection threshold. Thus, the size threshold for targets may lose its effectiveness in Local-CFAR- Th_{size} . For the proposed BMS method, which consists of Bayesian saliency map construction and morphological saliency map construction, we apply the size threshold to the Bayesian saliency map in morphological saliency map construction. The structures of the targets are complete and the sizes of the targets are reliable in the Bayesian saliency map, and thus we can apply the size threshold to the Bayesian saliency map via morphological saliency map construction to eliminate some too small or large clutter while retaining the targets.

In the following, we give the processing time for the SAR image shown in Fig. 3(a) by the proposed BMS method. The time cost of superpixel segmentation is 2072.1 s, Bayesian saliency map construction is 282.2 s, and morphological saliency map construction is 7.7 s, from which we can see that the superpixel segmentation and Bayesian saliency map construction cost a lot of time. Although the proposed BMS method is time-consuming under our current implementation, the speed of it can be accelerated via the parallel computing and code optimization.

IV. CONCLUSION

To overcome the drawbacks of the conventional target detection methods which are mainly based on the intensity differences between the targets and clutter, in this paper, we propose a new target detection method for high-resolution SAR images via BMS, which can make full use of the size prior information of the targets to be detected besides the intensity information of the targets. The major contributions of this paper are summarized as follows.

- 1) We develop a Bayesian saliency detection method for SAR images, which can obtain the smooth and complete structures of the bright objects in the SAR images.
- 2) We develop a morphological saliency detection method for SAR images, which constructs the morphological saliency map from the Bayesian saliency map instead of from the original SAR image, to highlight the targets of interest while suppressing the clutter.
- 3) We use not only the intensity information but also the size information of the targets for SAR target detection, and thus the proposed BMS method can eliminate the large man-made clutter with similar intensities to those of the targets of interest.

The quantitative and qualitative experimental results based on the challenging miniSAR real data set have verified that the proposed SAR target detection method can yield much better detection performance than the traditional target detection methods.

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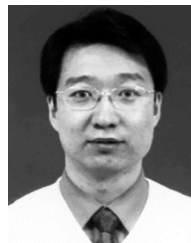


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